Practical 7: Tree and Graph Data Structures with DFS and BFS (CO2)

Code:

```
graph = {
    'A': ['B', 'C'],
    'B': ['D'],
    'C': ['D'],
    'D': []
}

def bfs(start):
    visited, queue, order = {start}, [start], []
    while queue:
        node = queue.pop(0)
        order.append(node)
        for n in graph[node]:
            if n not in visited:
                visited.add(n)
                queue.append(n)
    return order

def dfs(node, visited=None, order=None):
    if visited is None:
        visited, order = set(), []
    visited.add(node)
    order.append(node)
    for n in graph[node]:
        if n not in visited:
            dfs(n, visited, order)
    return order

print("BFS Traversal:", bfs('A'))
print("DFS Traversal:", dfs('A'))
```

Output:

```
BFS Traversal: ['A', 'B', 'C', 'D']
DFS Traversal: ['A', 'B', 'D', 'C']
```